

Cristina Rodríguez, PhD

Research facility leader · Translational imaging scientist · Research software builder

Preclinical Imaging Facility Manager, Molecular Imaging Research Facility (MIRF), The University of British Columbia · Vancouver, Canada

cristina.rodriguez@ubc.ca · cristinarod2.github.io · linkedin.com/in/crisrod · orcid.org/0000-0002-3313-4422

 12 yrs at UBC's imaging core; leading it since 2020	 17+ research groups supported	 5 partner institutions	 25 active projects coordinated	 5 modalities run across 3 UBC facilities	 60+ peer-reviewed papers
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Leadership profile

I am a scientific facility leader who connects translational imaging science with the operations, technology and institutional structure that make it work. I run UBC's small-animal in vivo imaging core, a PET, SPECT and CT facility that serves research groups across UBC, TRIUMF, SFU, BC Cancer and BC Children's Hospital, alongside its academic co-directors, Profs. Vesna Sossi and Urs Häfeli. I run its service model and day-to-day budget on a cost-recovery model, set and benchmark its pricing, manage its radiation-safety and animal-ethics compliance, report to institutional and scientific oversight, and lead the partnerships that bring studies in. I joined the lab in 2014 and helped build it into the facility I now run. Beyond the imaging facility itself, I have helped two other UBC laboratories establish optical imaging and microCT capabilities, so my operating experience spans five modalities across three facilities. I have co-taught PHAR 518 in the Faculty of Pharmaceutical Sciences since 2015 and lead the Faculty's optical imaging training.

My work combines three areas rarely held together in one role. A radiopharmaceutical scientist's judgement across the whole chain: from probe chemistry (PhD in medicinal and bioinorganic chemistry, 60+ peer-reviewed publications, from Alzheimer's radiotracer design to theranostic imaging with alpha-, beta- and Auger-emitting radionuclides), through scanner physics and quantitative analysis, to the pharmacokinetics and biodistribution that give images their meaning. An operator's discipline in budgets, compliance, governance and multi-partner delivery. And the ability to build the systems a facility needs, from protocols and training to the software that runs its daily operations.

Leadership experience

Preclinical Imaging Facility Manager, MIRF

Dec 2020 – present

Molecular Imaging Research Facility, The University of British Columbia, Centre for Comparative Medicine

STRATEGY AND OPERATING MODEL

- Define the service portfolio, pricing, capacity and strategic priorities of UBC's preclinical imaging core, serving 17+ research groups and 25 active projects in the latest programme report, built around a VECTor/CT system that acquires PET, SPECT and CT simultaneously, including dual-isotope imaging in the same animal.
- Lead financial planning and cost recovery for the facility: forecasting, pricing, cost measures and reporting. Rebuilt financial and activity reporting into a single ledger, and benchmarked the rate card against comparable Canadian facilities to give leadership evidence-based options.
- Led the successful applications for competitive infrastructure funding, including a \$145,000 NSERC Research Tools and Instruments award (2025) to upgrade core instrumentation and a UBC Research Facilities Support Grant (2020–2021), for a platform otherwise funded through cost recovery.

GOVERNANCE, COMPLIANCE AND OVERSIGHT

- Lead facility-level radiation-safety and animal-ethics compliance; report to institutional and scientific oversight bodies and serve as principal contact for TRIUMF's Life Sciences (LSPEC) programme across 25 projects.
- Manage the instrument lifecycle: quality control, calibration and cross-calibration programmes, service contracts and vendor relationships; author and version-control the facility's SOPs.

PARTNERSHIPS AND SCIENTIFIC LEADERSHIP

- Build and maintain the relationships that sustain the facility with investigators, departments and partner institutions; negotiate scope, timelines, responsibilities and cost for multi-year, multi-partner studies.
- Guide investigators from study design to interpretation and set the quantification methodology (calibrations, %ID/g, dosimetry) across diagnostic and therapeutic radionuclides, behind 63 facility publications since 2014, including first-in-field imaging with Ac-225, Hg-197m/g, Tb-155/161 and Er-165.

PEOPLE, TRAINING AND CAPACITY BUILDING

- Supervise and develop technical staff, students and facility users; lead the annual imaging workshops and one-to-one quantitative imaging training; co-instructor of PHAR 518 (Pharmaceutical Sciences) since 2015.
- Advised another UBC laboratory through establishing a new microCT capability in its first year (2024–2025): site set-up, workflow design and operating practice.
- Lead trainer and technical authority for optical imaging (Lago X) in the Pharmaceutical Sciences Analytical Suite since 2024: standardised acquisition and analysis protocols and onboarded researchers from multiple labs to one workflow.

OPERATIONAL TECHNOLOGY

- Designed and built LabTools, a software suite that is replacing the facility's manual workflows: tools for figure generation, quantitative analysis and animal transfers are in daily use, with SOP control and decay-corrected radioisotope inventory in development. Several tools are open source (cristinarod2.github.io/labtools) and used by other laboratories.

Lab Manager and Research Associate, Preclinical Nuclear Imaging

Oct 2014 – Dec 2020

The University of British Columbia

- Helped build UBC's preclinical PET, SPECT and CT platforms from its early years into the shared imaging service they are today, working with investigators across UBC, TRIUMF and local biotech: established instrument QC, calibration and quantification workflows, user training and study documentation.
- Worked hands-on across the full study chain: small-animal handling, anaesthesia, injection and surgical support under CCM protocols; radiotracer handling and dose preparation; VECTor/CT operation, reconstruction and analysis. This is why I can train and troubleshoot every role in the facility.

Beatriu de Pinós Postdoctoral Fellow, Medicinal Chemistry

Jul 2010 – Oct 2014

The University of British Columbia · independently funded; led the project

- Won independent funding and led a multi-year project on theranostic agents for Alzheimer's disease, including a drug-design collaboration with Pharmaceutical Sciences; supervised undergraduate and graduate students; Alzheimer's Drug Discovery Foundation Young Investigator Scholarship (2012).

Professora Ajudant (Assistant Lecturer) and PhD researcher

2005 – 2009

Universitat Autònoma de Barcelona, Spain

- Doctoral research in the UAB Department of Chemistry on radiotracers and multifunctional agents for Alzheimer's disease, from synthesis and chemoinformatic selection of drug-like candidates, through in vitro amyloid and metal-protein assays, to biodistribution in transgenic mouse models; research stays at ICMAB-CSIC (I-123 radiolabelling) and IQS (transthyretin amyloidosis); full-time teaching appointment throughout.

Scientific credentials

PhD, Medicinal and Bioinorganic Chemistry · Universitat Autònoma de Barcelona, 2009 · Excel·lent cum laude (highest distinction) · Multifunctional Agents as Alzheimer's Disease Theranostics

MSc, Chemistry · Universitat Autònoma de Barcelona, 2005 · Matrícula d'Honor (highest distinction) · metal-chelating agents for Alzheimer's disease

Publications · 60+ peer-reviewed papers (2007–2026), h-index 26, 2,500+ citations (Google Scholar, 2026), including J. Am. Chem. Soc., Angew. Chem., Nature Communications, ACS Nano and Physics in Medicine & Biology; corresponding author on three. Full list at cristinarod2.github.io and [ORCID](https://orcid.org/).

Recognition, service and certifications

Facility leadership · MILabs Image of the Year, Laboratory Award (2020) · World Molecular Imaging Congress MOMIL grant for leaders of small-animal imaging facilities (2019)

Research · Beatriu de Pinós Fellowship, one of four awarded in the Science panel (2010–2014) · Alzheimer's Drug Discovery Foundation Young Investigator Scholarship (2012) · ScienceDirect Top 25 most-downloaded article, Coordination Chemistry Reviews (2012)

Invited lectures · Keynote, CONFIQ-4, Mérida (2016) · Canadian Society for Pharmaceutical Sciences (2019) · Int. Laboratory Animal Technician Week (2020) · MBF and CDM Annual Meeting, UBC (2024)

Service · Organizing committee, International Conference on Magnetic Carriers, since 2015 (~330 participants) · peer reviewer for ACS and RSC journals · translator of Theodore Gray's The Elements into Catalan

Certifications · Graduate directed studies in pharmacokinetics, completed (DST 550, Pharmaceutical Sciences, 2020, Prof. A. Collier) · UBC radiation, chemical and biosafety · UBC Animal Care and CCM animal-user training · TDG, Transportation of Dangerous Goods (BCIT) · Languages: English, Spanish, Catalan (native)